

CHERNYAEVO, Russian Federation

WMO: 313710

Lat: 52.7896N

Lon: 126.0030E

Elev: 686

StdP: 14.34

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.2	-35.8	-44.5	0.4	-39.8	-40.4	0.6	-35.5	19.5	0.2	17.3	-3.9	0.3	50	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	86.9	69.0	83.4	67.4	80.3	66.1	72.2	81.6	70.3	79.2	68.5	76.6	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	110.6	76.5	67.4	103.1	74.1	65.7	97.2	72.3	36.5	81.8	34.8	79.2	33.3	76.7	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.5	13.5	-44.4	93.1	4.7	3.7	-47.8	95.8	-50.5	97.9	-53.1	100.0	-56.5	102.7		
(5)			-44.4	75.1	4.4	2.3	-47.5	76.8	-50.1	78.1	-52.5	79.5	-55.7	81.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.8	-13.6	-4.9	13.4	35.8	51.1	63.9	69.0	64.5	50.9	32.3	5.4	-12.2		
	DBStd	31.03	11.23	10.06	11.54	8.02	7.07	6.25	5.06	5.65	7.51	8.99	12.52	10.83		
	HDD50	9047	1972	1539	1136	431	72	1	0	1	78	552	1338	1927		
	HDD65	13114	2437	1959	1601	877	435	91	19	76	424	1015	1788	2392		
	CDD50	1677	0	0	0	5	105	419	590	450	106	2	0	0		
	CDD65	268	0	0	0	0	3	59	144	60	2	0	0	0		
	CDH74	2682	0	0	0	4	94	791	1205	552	36	0	0	0		
	CDH80	752	0	0	0	0	18	234	370	128	2	0	0	0		
	WSAvg	4.6	3.7	4.0	5.3	6.0	5.8	4.3	3.7	4.0	4.8	5.0	4.3	4.0		
	PrecAvg	19.1	0.2	0.2	0.4	1.2	1.7	2.9	4.8	3.7	2.6	0.7	0.5	0.3		
(15) Precipitation	PrecMax	30.0	1.0	0.8	0.9	3.3	5.0	6.9	8.3	9.3	6.2	2.2	1.3	0.9		
	PrecMin	11.7	0.0	0.0	0.0	0.3	0.3	0.7	0.9	1.3	0.5	0.0	0.1	0.0		
	PrecStd	3.6	0.2	0.2	0.2	0.6	1.1	1.7	1.8	2.0	1.4	0.6	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.1	27.5	48.4	71.2	82.9	91.2	92.7	88.1	78.3	63.2	36.8	21.8		
		MCWB	15.0	22.8	37.3	52.2	62.0	72.1	70.6	69.9	63.4	50.5	30.9	18.2		
	2%	DB	12.2	21.1	42.0	62.1	76.1	85.9	87.6	83.6	73.5	56.3	31.5	14.3		
		MCWB	9.9	18.1	33.0	46.8	57.0	67.2	70.2	68.6	60.8	46.0	27.8	12.4		
	5%	DB	8.1	16.6	37.0	56.1	71.3	82.3	84.0	79.9	68.9	51.1	27.9	8.2		
		MCWB	6.3	13.7	29.8	43.0	54.9	65.2	68.6	66.9	57.9	42.2	25.3	6.5		
	10%	DB	3.9	12.2	32.6	51.2	66.5	78.6	80.7	76.5	64.7	46.4	23.7	4.1		
		MCWB	2.4	9.7	27.3	40.0	52.2	63.9	67.7	65.5	55.2	38.6	21.0	2.8		
	0.4%	WB	15.2	22.9	38.1	52.3	63.2	74.7	75.6	73.0	65.6	52.1	31.8	18.6		
		MCDB	18.1	26.7	48.1	69.7	78.8	84.1	85.1	82.1	74.5	61.7	35.1	21.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	9.8	18.1	33.5	47.1	59.2	70.0	72.9	70.7	62.5	46.7	28.4	12.3		
		MCDB	12.2	21.2	40.9	60.3	72.9	82.3	81.6	80.0	70.9	55.7	31.3	14.4		
	5%	WB	6.3	13.8	30.5	43.5	56.4	67.3	71.2	68.8	60.0	42.9	25.2	6.4		
		MCDB	8.1	16.5	36.6	55.2	67.4	78.2	79.7	76.6	66.6	49.4	27.8	8.1		
	10%	WB	2.5	9.9	27.2	40.5	53.8	65.0	69.5	67.0	57.3	39.7	21.0	2.7		
		MCDB	4.0	12.1	32.5	49.7	64.3	75.5	77.6	74.0	63.0	45.8	23.5	4.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.8	26.9	26.7	22.9	23.2	22.5	19.1	20.2	22.0	19.4	19.5	18.5		
		MCDBR	19.6	29.1	27.7	33.0	33.5	29.5	26.4	26.4	28.6	27.7	20.8	17.8		
		MCWBR	17.8	25.8	19.9	19.5	17.1	13.0	11.0	12.6	15.7	17.9	17.0	16.3		
		MCDBR	19.3	29.2	26.5	31.2	28.1	24.8	20.3	21.3	23.8	25.6	20.8	17.7		
	5% WB	MCWBR	17.6	26.1	19.4	18.8	15.2	12.2	9.9	11.1	14.0	17.2	17.0	16.3		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.293	0.360	0.535	0.593	0.524	0.484	0.421	0.387	0.369	0.303	0.265		
	taud		2.144	2.086	1.955	1.700	1.670	1.932	2.103	2.274	2.283	2.162	2.129	2.084		
	Ebn at Noon		227	253	254	220	215	235	243	254	248	220	202	196		
	Edh at Noon		22	32	45	67	73	56	47	37	33	30	22	19		
(44) All-Sky Solar Radiation	RadAvg		429	798	1346	1674	1685	1925	1683	1489	1149	744	461	328		
	RadStd		16	22	52	159	124	193	161	115	90	49	23	16		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page